true metric

# Invariant to rigid transformations

# Gromov-Wasserstein: Properties

- It defines a **true metric** on the space of metric–measure spaces (up to isometry):
  1. $\mathrm{GW}((\alpha, d_X), (\beta, d_Y)) \geq 0$, and =0 iff the metric–measure spaces are isometric (i.e. ∃ bijection $\phi : X \to Y$ that satisfies $\phi_\sharp \alpha = \beta$, $d_X(x, x') = d_Y(\phi(x), \phi(x'))$
  2. Symmetric: $\mathrm{GW}((\alpha, d_X), (\beta, d_Y)) = \mathrm{GW}((\beta, d_Y), (\alpha, d_X))$
  3. Satisfies the triangle inequality
- **Invariant to rigid transformations** (translation, rotation, scaling if distances rescaled accordingly).
- Admits a geodesic structure (Sturm 2012): allows defining gradient flows and barycenters of metric–measure spaces.

# Gromov-Wasserstein: Computation